

2-DOF Helicopter Control Via State Feedback and Full/Reduced-Order Observers

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Abstract—This paper presents an analysis of state feedback control and state estimation applied to the two degrees-of-freedom (2-DOF) helicopter platform Quanser Aero 2 to achieve trajectory tracking control and stabilization. It involves establishing a dynamic model of the Quanser Aero 2 platform using the Euler-Lagrange equations of motion to obtain a MIMO system with four equations of state and two inputs. Then, the design of two state feedback controllers using full and reduced order observers, respectively, to stabilize the Aero 2 system. In practice, not all system states can be directly measured, which affects the control system. Observers can estimate these unmeasured states based on the available measurements and the system's model. Therefore, controllers with observers are implemented in real-time on the Quanser Aero 2 platform and compared to an optimal state feedback control, namely, the linear quadratic regulator (LQR). Experimental results are presented and compared to analyze the performance of 2-DOF helicopter control via state feedback and observer estimation.

Index Terms—State feedback control, full-order observer, reduced-order observer, LQR.

I. INTRODUCTION

The use of Unmanned Aerial Vehicles (UAVs) has witnessed a remarkable rise in recent years. This is due to their adaptability and the multitude of applications they offer across various fields, such as surveillance, search and rescue missions, and scientific research [1], [2]. Among the different types of UAVs, helicopters offer unique advantages, such as the ability to hover and perform precise maneuvers in confined spaces. However, the control of helicopter dynamics poses significant challenges due to their inherent nonlinearities and coupled dynamics.

This study focuses on the control problem of the Aero 2, a two-degrees-of-freedom (2 DOF) helicopter experimental setup. The Aero 2 is a widely used control education and research platform, providing a simplified yet representative model of a real helicopter's dynamics [3]. This work aims to design and implement feedback control strategies to stabilize the Aero 2 and achieve precise tracking of desired trajectories.

State feedback control is a powerful method that directly employs the system's state variables to calculate the control input. However, in many practical applications, not all state variables are directly measurable or available from sensors [4], [5]. This is where state observers come into play. State

observers are model-based estimation algorithms that reconstruct the system's states from the available measurements and input signals. By incorporating state observers into the control loop, state feedback control can be implemented even when some states are not directly measurable.

In this study, we investigate two types of state observers: full-order observers and reduced-order observers [6], [7]. Full-order observers estimate all the system states, providing a complete representation of the system dynamics. While accurate, these observers may be computationally expensive and sensitive to model uncertainties. In contrast, reduced-order observers estimate only a subset of the states, potentially reducing computational complexity and increasing robustness to model inaccuracies [8]. However, this reduction in observer order may come at the cost of decreased estimation performance or stability margins. The choice between full-order and reduced-order observers involves trade-offs between estimation accuracy, computational requirements, and robustness considerations specific to the application.

We explore three control approaches: linear quadratic regulator (LQR) control, state feedback with a full-order observer, and state feedback with a reduced-order observer. The LQR controller is designed to minimize a quadratic cost function that balances control effort and state deviations [9]. On the other hand, the state feedback controllers rely on estimates of the system states obtained through observers, either full-order or reduced-order, to determine the control inputs.

The performance of these three control strategies is evaluated in terms of their ability to stabilize the Aero 2, and track desired trajectories accurately. Hardware experimental results are presented to assess the effectiveness of each approach under various operating conditions and disturbances. The contributions of this work are twofold. First, we provide a comprehensive comparison of the three control strategies for the Aero 2 system, highlighting their respective strengths and limitations. Second, we demonstrate the successful implementation of these control approaches on a real experimental platform, addressing practical challenges such as sensor noise and model uncertainties.



Fig. 1. Quanser Aero 2 Helicopter configuration

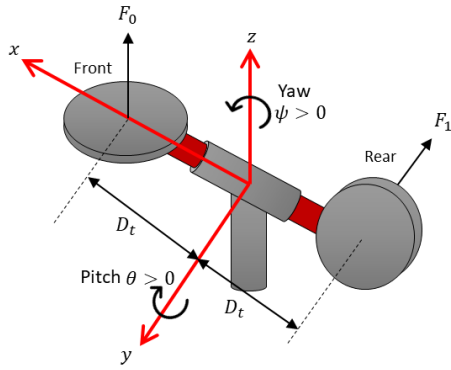


Fig. 2. Free body diagram of the Aero 2 [10]

II. SYSTEM DESCRIPTION AND MODELING

Currently, the control laboratory of the Center of Development of Advanced Technologies (CDTA) has a control system Aero 2 of the company Quanser [10]. This system is shown in Fig. 1. It is a two-rotor laboratory apparatus for flight control-based studies. The system can be set up as a helicopter-like configuration with the primary thruster in a horizontal position and the tail thruster in a vertical position, both powered by two DC motors.

The system consists of a compact base unit featuring an integrated amplifier with current-sensing capabilities and a built-in data acquisition device for collecting measurements. A versatile QFLEX 2 interface panel offers connection options to devices such as personal computers, embedded computers, and microcontrollers [10]. The system utilizes four high-resolution optical encoders along with an Inertial Measurement Unit (IMU) for precise attitude measurement and control in both pitch and yaw axes. Notably, slip ring wiring facilitates unrestricted, continuous 360-degree yaw rotation.

When the pitch motor receives a voltage V_p , the front rotor produces a force that acts normally on the body away from the pitch axis (along the x-axis). Therefore, the torque produced by the rotation of the front propeller blade also produces a torque about the yaw-axis because of aerodynamic drag (z-axis). Because of this, traditional helicopters have a tail rotor

to balance out the torque the big main rotor creates around the yaw axis. Similar to the front motor, the rear motor produces a force that affects the body away from the yaw axis.

The equations of motion of the Aero 2 are expressed from the free body diagram (Fig. 2) about the horizontal axis as [10]:

$$\begin{aligned} J_p \ddot{\theta} + D_p \dot{\theta} + K_{sp} \theta &= \tau_p, \\ J_y \ddot{\psi} + D_y \dot{\psi} &= \tau_y \end{aligned} \quad (1)$$

Where the pitch and yaw torques are given by:

$$\begin{aligned} \tau_p &= K_{pp} D_t V_p + K_{py} D_t V_y, \text{ and} \\ \tau_y &= K_{yp} D_t V_p + K_{yy} D_t V_y. \end{aligned} \quad (2)$$

The parameters used in the (1) and (2) are listed in Table. I.

TABLE I
PARAMETERS OF THE QUANSER AERO 2.

Symbol	Description	Value
J_p	Pitch axis inertia	0.0232 $Kg.m^2$
J_y	Yaw axis inertia	0.0238 $Kg.m^2$
D_p	Pitch axis damping	0.0020 $N.m/V$
D_y	Yaw axis damping	0.0019 $N.m/V$
K_{sp}	Pitch axis stiffness	0.0074 $N - m/V$
K_{pp}	Pitch thrust gain from front rotor	0.0032 N/V
K_{py}	Pitch thrust gain from rear rotor	0.0014 N/V
K_{yy}	Yaw thrust gain from rear rotor	0.0061 N/V
K_{yp}	Yaw thrust gain from front rotor	-0.0032 N/V
D_t	Distance between pivot and rotor center	0.1674 m

The Aero 2 User Manual [10] provides some of the model parameters. The remaining parameters are determined experimentally through identification techniques. Using the state variables vector $x^T = [\theta(t), \psi(t), \dot{\theta}(t), \dot{\psi}(t)]$, representing the pitch angle, yaw angle, pitch velocity and yaw velocity respectively, the following linear state-space representation of the Aero 2 system is obtained:

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned} \quad (3)$$

where

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -K_{sp}/J_p & 0 & D_p/J_p & 0 \\ 0 & -K_{sy}/J_y & 0 & -D_y/J_y \end{bmatrix} \text{ and}$$

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ D_t K_{pp}/J_p & D_t K_{py}/J_p \\ D_t K_{yp}/J_y & D_t K_{yy}/J_y \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \text{ and } D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Equation (3) represents a multiple-input, multiple-output (MIMO) system with two outputs $y^T = [\theta(t), \psi(t)]$ and two inputs $u^T = [V_p(t), V_y(t)]$. Since the system starts when it is at rest, all the initial conditions are zero.

III. STATE FEEDBACK CONTROL WITH OBSERVERS

The control objective is to generate torques that allow for accurate trajectory tracking of pitch and yaw angles to desired ones with the smallest errors. Let us consider the desired state vector as trajectories as $x_d^T = [\theta_d(t), \psi_d(t), 0, 0]$.

State feedback control is a method that offers significant advantages and simplicity in control design. It assumes access to all system states for feedback (full-state feedback) [4], [11]. This allows us to directly incorporate these states into the control law while using a gain matrix (K) to adjust the system input based on the current state of the system. This method is implemented using MATLAB functions that generate the gain matrix based on (3). The resulting gain matrix is then used to calculate the system input, which in turn influences the system's behavior.

$$u(t) = -K(x_d^T - x^T) \quad (4)$$

where K is a constant state feedback gain matrix of dimension 2×4 in the case of the Aero 2 system.

Full-state feedback controllers, while theoretically advantageous, face practical challenges. Firstly, obtaining measurements for all state variables can be difficult and expensive, especially in large-scale systems where numerous feedback loops are required. Secondly, not all state variables may be directly measurable in practice. Instead, we often rely on output signals, which are linear combinations of the true state variables. For control purposes, this incomplete state knowledge is a major drawback, as it makes it impossible to control these plants satisfactorily in such cases. To address this challenge, the concept of an observer is employed.

A. Full order observer

The state observer's function is to adjust the information in the system's states. This information may be directly linked to the states of the system [11]. The full-order observer estimates all the states of the system and compares the theoretical system model to the physical model, analyzing the error using a gain matrix (L) that multiplies the difference between y and $\hat{y}$. The matrix L is then incorporated into the theoretical model and is adjusted until the error becomes practically zero. Consequently, the full-order observer's output provides estimated states that can be used for system feedback.

After checking the observability of our state space model ((3)), we design the full-order observer as [6],

$$\begin{aligned} \dot{\hat{x}}(t) &= A\hat{x}(t) + Bu(t), \\ \hat{x}(t_0) &= \hat{x}_0, \\ \hat{y}(t) &= C\hat{x}(t). \end{aligned} \quad (5)$$

The error $e(t) = x(t) - \hat{x}(t)$ is used to construct the observation system.

$$\begin{aligned} \dot{\hat{x}}(t) &= A\hat{x}(t) + Bu(t) + L(y(t) - \hat{y}(t)) \\ &= A\hat{x}(t) + Bu(t) + LCe(t), \end{aligned} \quad (6)$$

where matrix L represents the observer gain matrix. From (3) and (6), we obtain the dynamics of the error as:

$$\dot{e}(t) = (A - LC)e(t), \quad (7)$$

In designing the observer, selecting the matrix L that ensures asymptotic stability (eigenvalues of $(A - LC)$ with negative real parts) guarantees that the estimation error, $e(t)$, converges to zero for any initial condition [12].

It is important to note that the observer dynamics should be significantly faster than those of the system itself. This is particularly crucial when the estimated state variables are used for feedback control purposes.

B. Reduced-order observer

Reduced-order observers present an attractive solution for state estimation in systems with high model order or computational constraints [7]. By estimating only a subset of the system states, rather than the full state vector, reduced-order observers can potentially reduce the computational burden and increase robustness to model uncertainties [6]. Reduced order observers should be used for systems that are not too noisy [13].

We consider the linear system defined by (3) with available measurements. To design a reduced-order observer, we assume that pitch and yaw angles are measurable, and estimate pitch and yaw velocities. An arbitrary matrix C_1 of dimension 2×4 whose rank is equal to 2 is used to construct an augmented matrix with full rank [6]:

$$\text{rank} \begin{bmatrix} C \\ C_1 \end{bmatrix} = 4, \quad (8)$$

Then, we introduce a vector $h(t)$ such that:

$$\begin{bmatrix} y(t) \\ h(t) \end{bmatrix} = \begin{bmatrix} C \\ C_1 \end{bmatrix} x(t), \quad (9)$$

Yields,

$$x(t) = \begin{bmatrix} C \\ C_1 \end{bmatrix}^{-1} \begin{bmatrix} y(t) \\ h(t) \end{bmatrix} = [L_1 \quad L_2] \begin{bmatrix} y(t) \\ h(t) \end{bmatrix} = L_1 y(t) + L_2 h(t). \quad (10)$$

The estimation of states of the Aero 2 is given by:

$$\hat{x}(t) = L_1 y(t) + L_2 \hat{h}(t), \quad (11)$$

While the system measurements, $y(t)$, are readily available, the state vector, $h(t)$, requires estimation using a reduced-order observer with dimension r ($r < n$). To initiate the observer design process for $h(t)$, we use the procedure described in [6], which is presented in the sequel.

$$\begin{aligned} \dot{h}(t) &= \frac{d}{dt} \{h(t)\} = \frac{d}{dt} \{C_1 x(t)\} = C_1 \dot{x}(t) \\ &= C_1 A x(t) + C_1 B u(t) = C_1 A L_2 h(t) + C_1 B u(t). \end{aligned} \quad (12)$$

Its time derivative $\dot{\hat{h}}(t)$ is given by:

$$\dot{\hat{h}}(t) = C_1 A L_2 \hat{h}(t) + C_1 A L_1 y(t) + C_1 B u(t) + K_1 (\dot{y}(t) - \dot{\hat{y}}(t)), \quad (13)$$

The reduced-order observer gain, denoted by K_1 , is chosen to drive the observer error, $h(t) - \hat{h}(t)$, to zero as time progresses.

Using system output, we can obtain more information about $y(t)$ from the knowledge of $\dot{y}(t)$, such as:

$$\begin{aligned}\dot{y}(t) &= C\dot{x}(t) = CAx(t) + CBu(t) \\ &= CAL_1p(t) + CALy(t) + CBu(t).\end{aligned}\quad (14)$$

And its estimate is given by :

$$\dot{\hat{y}}(t) = CAL_2\hat{h}(t) + CALy(t) + CBu(t). \quad (15)$$

Replacing (15) into (13), gives:

$$\begin{aligned}\dot{\hat{p}}(t) &= C_1AL_1\hat{p}(t) + C_1ALy(t) + C_1Bu(t) \\ &+ K_1(\dot{y}(t) - CAL_1\hat{p}(t) + CALy(t) + CBu(t)).\end{aligned}\quad (16)$$

Moving the term containing $\dot{y}(t)$ to the left side of (16) and implementing a variable transformation.

$$\hat{f}(t) = \hat{h}(t) - K_1y(t), \quad (17)$$

Which implies $\dot{\hat{f}}(t) = \dot{\hat{h}}(t) - K_1\dot{y}(t)$,

Therefore, the reduced-order observer for $\hat{f}(t)$ is:

$$\dot{\hat{f}}(t) = A_f\hat{f}(t) + B_fu(t) + K_fy(t), \quad (18)$$

Where:

$$\begin{aligned}A_f &= (C_1 - K_1C)AL_1, \\ B_f &= (C_1 - K_1C)B, \\ K_f &= (C_1 - K_1C)A(L_1 + L_2K_1).\end{aligned}\quad (19)$$

Using the observation for $\hat{f}(t)$ obtained from (18) and (19), the reduced-order observer estimate is derived from (17) as $\hat{p}(t) = \hat{f}(t) + K_1y(t)$, resulting in the following estimation of the original state-space variables.

$$\begin{aligned}\hat{x}(t) &= L_1y(t) + L_2\hat{h}(t) = L_1y(t) + L_2(\hat{f}(t) + K_1y(t)) \\ &= (L_1 + L_2K_1)y(t) + L_2\hat{f}(t).\end{aligned}\quad (20)$$

IV. REAL-TIME IMPLEMENTATION RESULTS

In this section, state feedback controllers with full and reduced order observers are hardware implemented on the Quanser Aero 2 helicopter and compared to optimal state feedback control, namely, the Linear Quadratic Regulator (LQR) [9]. Initial conditions are considered null $x_0 = 0$. The experimental results using square input waves as input signals for pitch and yaw angles are depicted in Fig. 3 and Fig. 4, whereas control inputs are described in Fig. 5 and Fig. 6.

The depicted control strategies showed distinct performance for pitch and yaw. For pitch tracking, the full-order observer controller excelled, following the desired trajectory closely and aligning with optimal LQR control. This likely stems from its ability to accurately estimate all states, crucial for

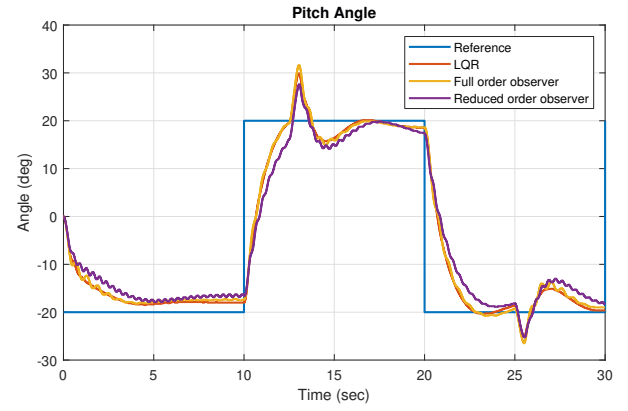


Fig. 3. Pitch tracking using feedback control and state estimation

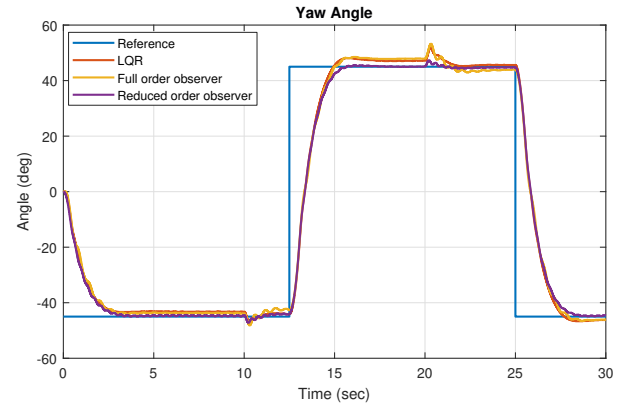


Fig. 4. Yaw tracking using feedback control and state estimation

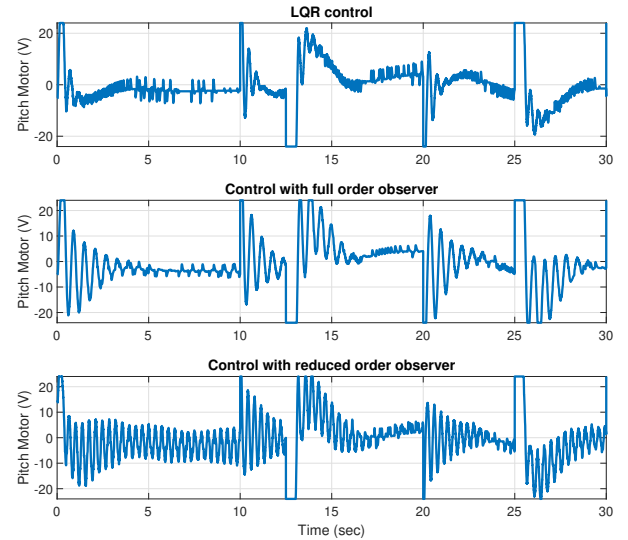


Fig. 5. Pitch motor input

precise pitch control. Conversely, the reduced-order observer controller achieved superior yaw tracking. This might be because yaw dynamics are simpler, requiring fewer states for control. The reduced-order observer is then sufficient for

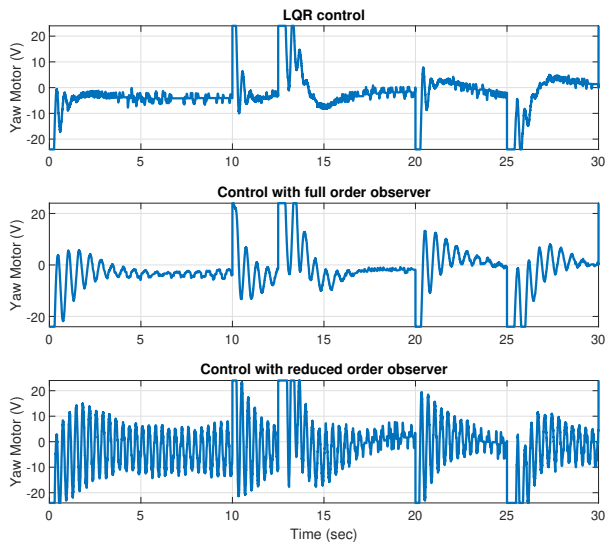


Fig. 6. Yaw motor input

yaw, while a full-order observer might introduce unnecessary complexity.

The analysis of motor control inputs reveals an interesting observation. While both state feedback approaches utilize more control effort compared to LQR, the state feedback with a reduced-order observer exhibits the highest control input demands for both pitch and yaw. This could be attributed to the trade-off between accuracy and efficiency. The reduced-order observer might introduce a slight estimation error that necessitates a stronger control input to achieve the desired tracking performance.

The estimated states using both the full order observer and the reduced order observer are depicted in Fig. 7 and Fig. 8.

To evaluate the robustness of the state feedback controllers with observers against unknown external disturbances, we conduct stabilization experiments by fixing the pitch angle to 15° and yaw angle to 30° and introduce a random disturbance by hand on the Aero 2 after 15 sec. The obtained results are

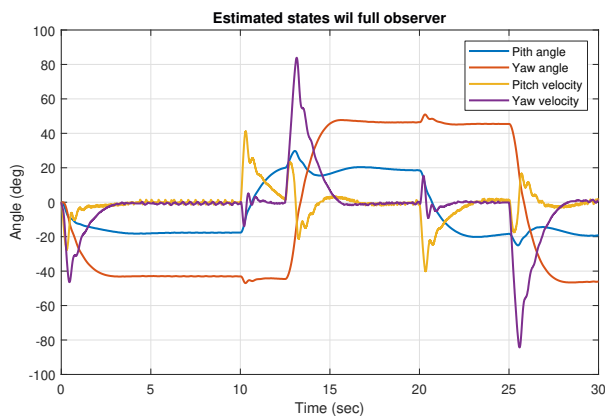


Fig. 7. Estimated states using full order observer

illustrated in Fig. 9 to Fig. 12.

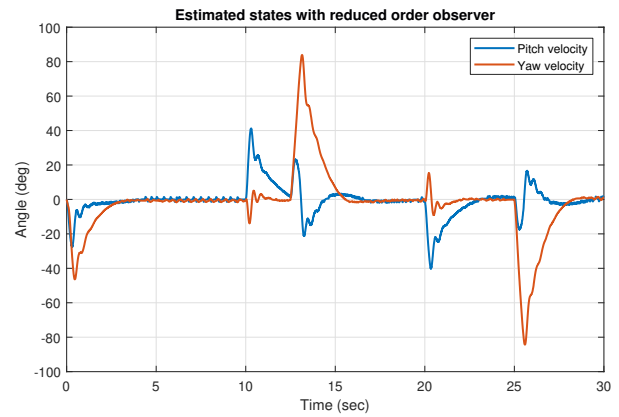


Fig. 8. Estimated states using reduced order observer

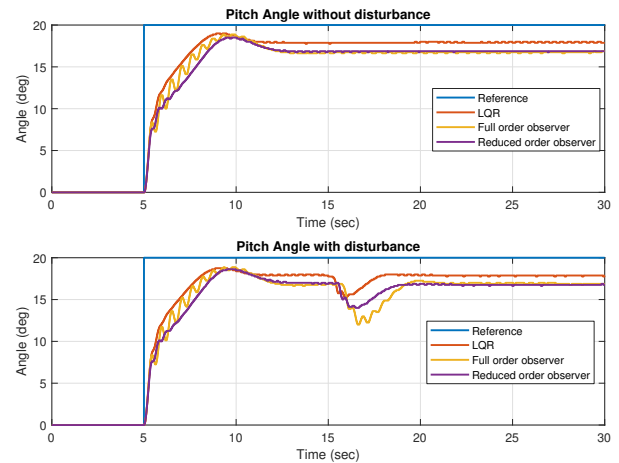


Fig. 9. Pitch stabilization with and without disturbance

For pitch control, all three approaches achieved stabilization, but a persistent steady-state error remained. This error was minimized by the LQR controller. The state feedback controller with a full-order observer initially displayed oscillations, possibly due to state estimation error. However, both state feedback controllers recovered well from disturbances, highlighting the pitch's higher sensitivity which likely arises from its direct impact on longitudinal movement and stability.

In contrast, yaw control achieved no steady-state error. Furthermore, both LQR and the reduced-order observer controller demonstrated the best stabilization results, even under disturbances, as compared to state feedback with the full-order observer. This suggests yaw dynamics are less critical for overall stability compared to pitch.

For control inputs in case of stabilization with and without disturbances, the control effort by LQR is optimal, followed by state feedback with full order observation and then state feedback with reduced order estimation. we can observe an increase at the moment where disturbances are introduced, to counteract the effects of these external unknown disturbances and keep stability of the system.

Control input analysis of Fig. 11 and Fig. 12 revealed

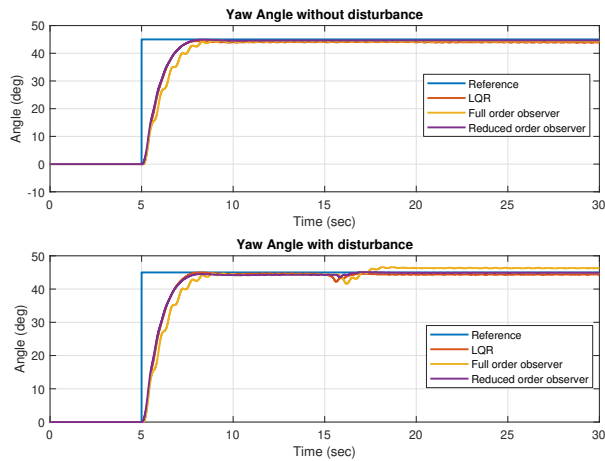


Fig. 10. Yaw stabilization with and without disturbance

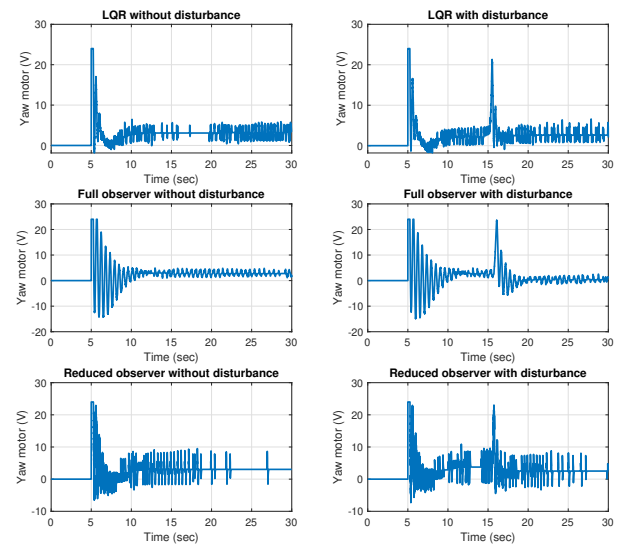


Fig. 12. Yaw motor input for stabilization

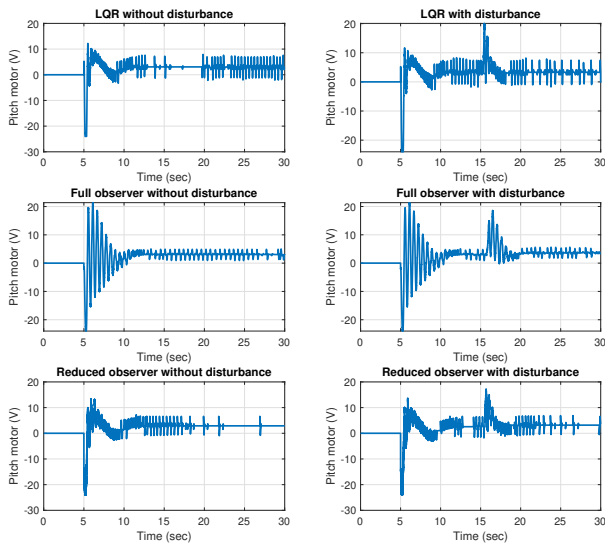


Fig. 11. Pitch motor input for stabilization

that LQR control minimized the effort in all scenarios (with and without disturbances), demonstrating its efficiency. While state feedback controllers using full-order and reduced-order observers required slightly higher control effort, particularly when counteracting external disturbances, their performance still closely approached the optimal control achieved by LQR. This suggests that even with observers, we can achieve near-optimal control for both pitch and yaw axes, with all approaches exhibiting increased control effort during disturbances to maintain system stability.

V. CONCLUSION

In this study, we presented the design and implementation of two state feedback controllers with full and reduced-order observers and their comparison with optimal control LQR. Observers are important in control systems for state estimation, improved control performance, and reduced sensor requirements. These controllers were compared experimentally

to the tracking and stabilization of the Quanser Aero 2 helicopter system. Experimental results revealed that LQR control offers the most efficient approach, while state feedback with observers provides a valuable alternative for achieving near-optimal performance with slightly increased control effort. This is particularly beneficial for systems with complex or uncertain dynamics. Future research could explore adaptive control strategies that can adjust parameters in real-time to address varying operating conditions and disturbances.

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